

PreLab1 Report

Table No. 8 Name & Stu No. 2025531083 李忱昊 Score _____ of 100

PART ONE Lab Safety Rules and Regulations

Read and understand the Laboratory Rules and Regulations documents available on Blackboard. By signing below, I affirm that:

- [1] I have understood the Laboratory Rules and Regulations.
- [2] I agree to comply fully with all Laboratory Rules and Regulations.
- [3] I understand that any violation of the Laboratory Rules and Regulations may affect my final course grade.

Signature: 李忱昊

Date: 2026.3.14

PART TWO

1. Read the "Equipment Tutorial" to acquaint yourself with the basic equipment.
2. Thoroughly review the Lab 1 guide.

PART THREE

Breadboard is used to make up temporary circuits for testing or to try out an idea. No soldering is required so it is easy to change connections and replace components. Breadboards come in various brands and sizes to suit the requirements of the electronic circuits. Nevertheless, all breadboards share the same fundamentals.

1. Read the document about using breadboards.
2. Connections on breadboard

Breadboards have many tiny sockets (called 'holes') in it and strips of metal below the board which connect the holes placed on the top of the board. All "holes" that are electrically connected are said to share a common node.

Use the breadboards shown in Figure 1 as a model for your bread board. Draw lines connecting "holes" which are electrically connected. (6pts, 3pts each)

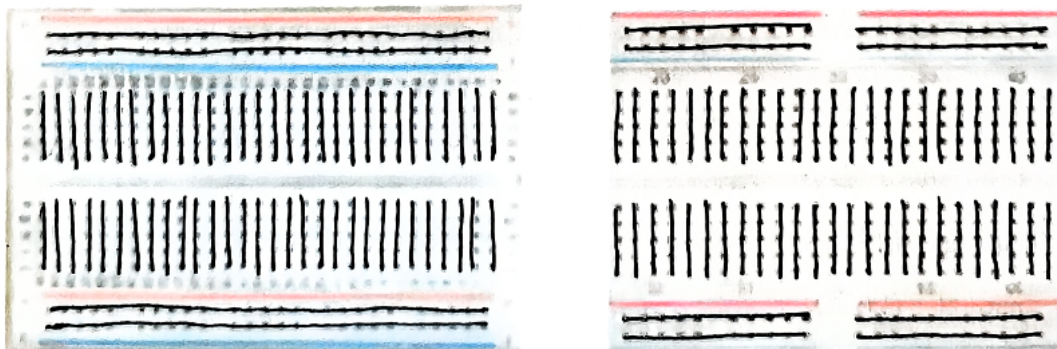
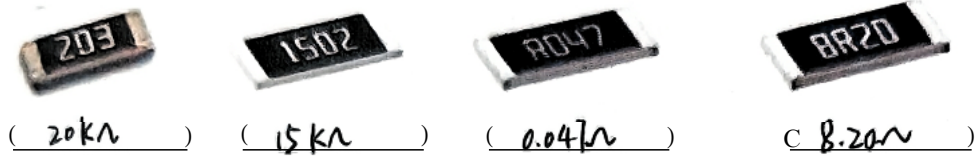


Figure 1

PART FOUR

1.

1) Read the value of those chip resistors below.(4 pts. 1pt each)



2.

1) For a 1k Ω resistor, what are the color band colors and associated band values?(5 pts. 1pt each)

Table 1

Resistor Value	Band1	Band2	Band3	Band4	Band5
1 k Ω , 1%	brown	black	black	brown	brown

What is the expected nominal resistance and tolerance (in Ohms)? (4 pts. 1 pt each)

R = . 1000 Ω $\pm$ 10 Ω (not%)Rmin = . 990 Ω Rmax = 1010 Ω

3. Given the color band sequence 'brown-green-black-brown-brown' in Figure 2,



Figure 2

1) What is the value of this resistor R? (2pts. Make sure to include units!)

1500 Ω

2) What is the value of tolerance? What are the largest and smallest values this resistor can have?(6 pts, 2 pts each)

tolerance: $\pm 1\%$ largest: 1485 Ω smallest: 1515 Ω

4. Given Determine the maximum current that a $470\ \Omega$, $\frac{1}{4}\text{W}$ resistor can have flowing through it without exceeding its power rating.(2 pts)

$$I_{\max} = \sqrt{\frac{P_{\max}}{R}} = \sqrt{\frac{\frac{1}{4}}{470}} \approx 0.023\text{ A}$$

5. Determine the maximum voltage that a $470\ \Omega$, $\frac{1}{4}\text{ W}$ resistor can have across it without exceeding its power rating.(2 pts)

$$U_{\max} = \sqrt{P_{\max} R} = \sqrt{\frac{1}{4} \times 470} \approx 10.840\text{ V}$$

PART FOUR

1.

- 1) Read the document about using Digital Multimeter.

Appendix7_ Fluke Multimeter user guide

The following questions pertain to the Digital Multimeter Fluke 15B+ used in the lab.

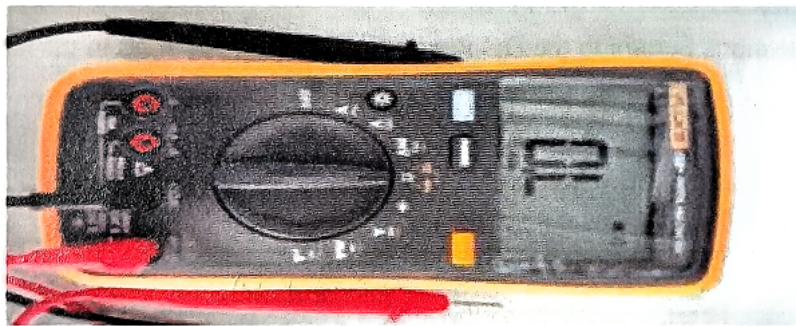


Figure 3 Digital Multimeter Fluke 15B+

- 2) What is a Digital Multimeter? (3 pts)

A DMM is a electronic measuring instrument used to measure current, voltage, resistance, capacitance, etc. with ~~high~~ precision.

- 3) There are four input terminals on the front of the device. Search for the documents you received or surf on the Internet, in your own words describe what each of these four input terminals are generally used for? (4 pts, 1pt each)

1. com. common ground, used as the ground reference.
2. 10A used to measure for large current measurement.
3. mA, μA for small current measurement.
4. 'V, Ω , 'C': for measuring voltage, resistance, capacitance.

- 4) How should a digital multimeter be connected in a circuit when measuring current:
in series or in parallel?(2pts)

in series

- 5) How should a digital multimeter be connected in a circuit when measuring voltage:
in series or in parallel?(2 pts)

in parallel

PART FIVE

1. Read the document about using DC Power Supply.
2. What does a DC power supply do?(3 pts)

A DC power supply provides a constant ~~not~~ voltage (with respect to time) for powering electronic circuits.

The following questions pertain to the DC Power Supply DH1766 used in the lab.

3. Complete the table below. (3 pts, 1pt each)

Table 2

Output Voltage Rang(reference to GND)	Value
CH1	<i>0~32V, 5A</i>
CH2	<i>0~32V, 5A</i>
CH3	<i>0~6V, 3A</i>

4. The output voltage of the DC power supply (CH3) is set to 1.5volts. Use the DMM to measure the output voltage of the DC power supply. Sketch the Power Supply and the DMM FLUKE 15B+, including the position the rotary switch should be, the connection of the red test lead and the black test lead. (3 pts)

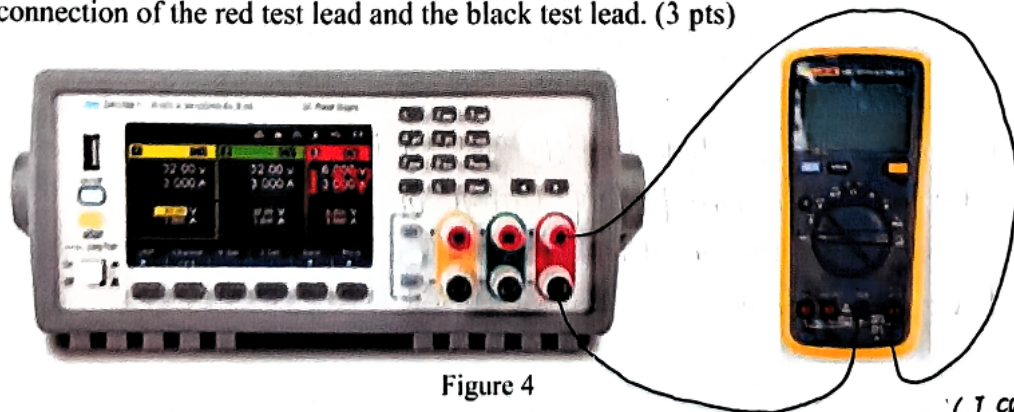
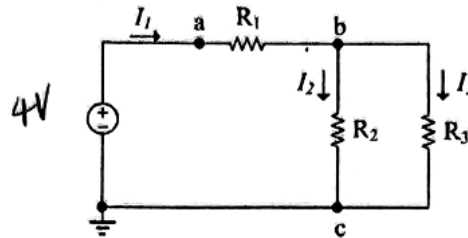


Figure 4

I can't see the picture clearly, so I finish this complete question based on a picture on the Internet)

PART SIX

For the circuit shown in Figure 5, assume $R_1=1.5\text{k}\Omega$, $R_2=R_3=1\text{k}\Omega$.

**Figure 5**

1.

- 1) Use theoretical analysis to determine the expected current and record the values in Table3(4pts). What equation did you use? Show your work.(4pts)

$$R_{\text{out}} = R_1 + R_2 \parallel R_3 = R_1 + \frac{R_2 R_3}{R_2 + R_3} = 1.5\text{k} + \frac{1\text{k} \times 1\text{k}}{2\text{k}} = 2\text{k}\Omega$$

$$I_1 = \frac{V}{R_{\text{out}}} = \frac{4}{2\text{k}\Omega} = 2 \times 10^{-3} \text{A} = 2\text{mA}$$

$$I_2 = I_3 = \frac{1}{2} I_1 = 1\text{mA}$$

Table 3

I_1 Theory(mA)	I_2 Theory(mA)	I_3 Theory(mA)	$I_2 + I_3$ Theory(mA)
2	1	1	2

- 2) Using the theoretical current found above, apply Ohm's law to determine the expected voltage drops across R_1 , R_2 , and R_3 . What equation did you use? Show your work(4pts). Record the values in Table 4(4pts).

$$V_{ab} = I_1 R_1 = 1.5\text{V}$$

$$V_{bc} = V - V_{ab} = 2.5\text{V}$$

Table 4

	$V_{ab}(\text{V})$	$V_{bc}(\text{V})$	$V_{ab} + V_{bc}(\text{V})$	$V_S(\text{V})$
1 k Ω , 1 %	1.5	2.5	4	4

2. In Figure 6, try to connect the resistors in series to the DC Power supply to make a complete circuit as shown in Figure 5. (6 pts)

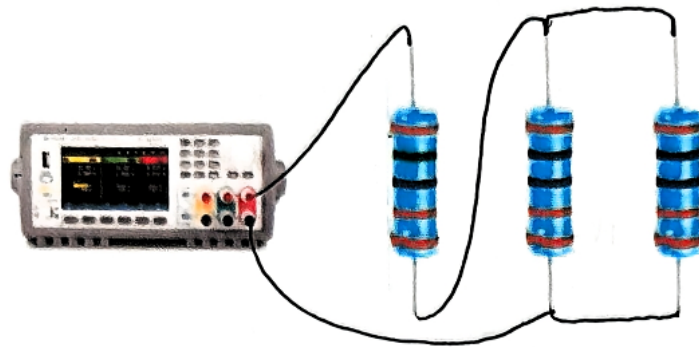


Figure 6

3. In Figure 7, sketch the connection to measure the voltage V_{ab} in Figure 5, including the position where the rotary switch should be. (6 pts)

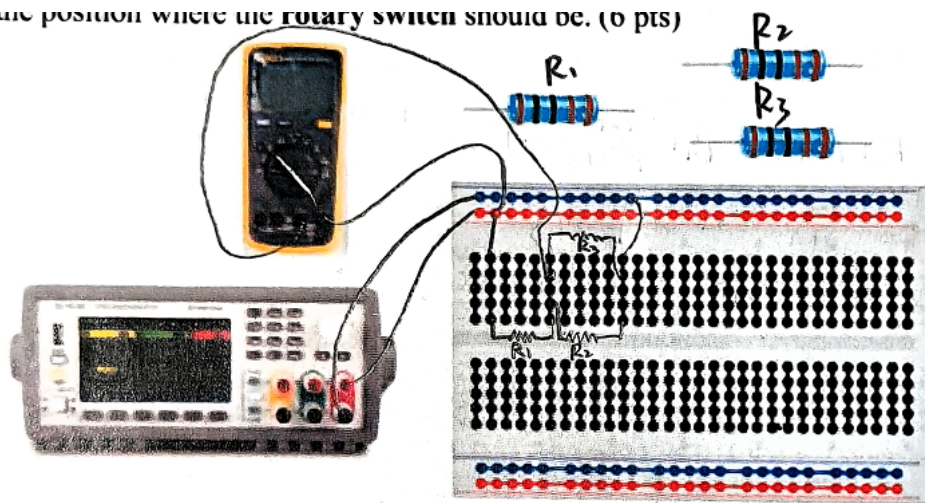


Figure 7

4. Sketch the connection in Figure 8 to measure the current I_2 in Figure 5, including the position where the rotary switch should be. (7 pts)

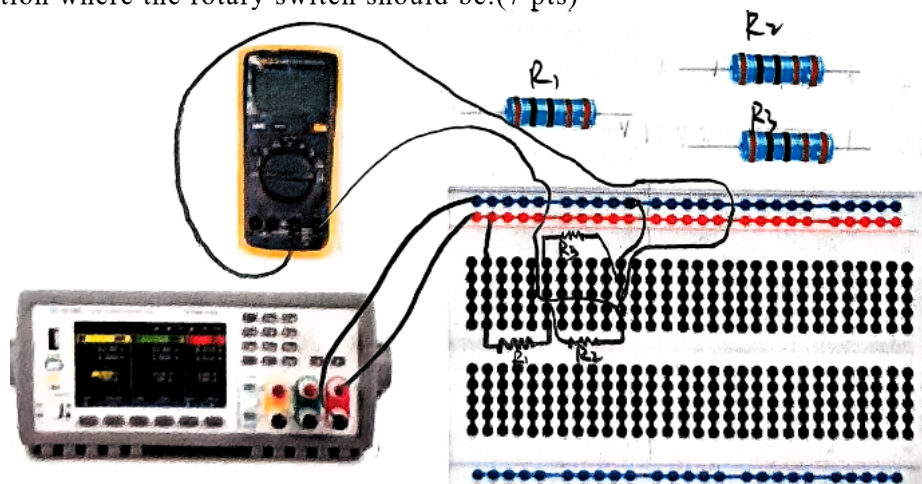


Figure 8

5. An alternative to using Ohm's law is to use a voltage divider rule. A voltage divider is a circuit that produces an output voltage (V_{out}) that is a fraction of its input voltage (V_{in}). A simple voltage divider is shown in figure 9. Note that in figure 9, the reference voltage is ground. In this circuit,

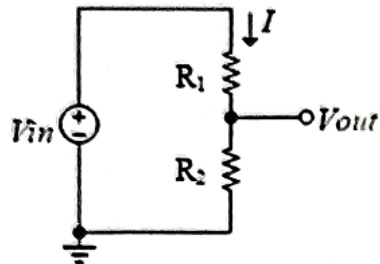


figure 9

$$V_{out} = \left(\frac{R_2}{R_1 + R_2} \right) \times V_{in}$$

In Figure 5, What is the voltage divider equation for the voltage across R_1 ? (2 pts)

$$\begin{aligned} V_{in} &= V_{in} = 4 \text{ V} \\ V_{out} &= \frac{R_2 \parallel R_3}{R_1 + R_2 \parallel R_3} \times V_{in} = 4 \times \frac{0.5}{2} = 1 \text{ V} \\ \therefore V_{R_1} &= V_{in} - V_{out} = 4 - 1 = 3 \text{ V} \end{aligned}$$

6. A potentiometer (variable resistor), or pot for short is a three-terminal resistive device. The schematic symbol for a potentiometer is shown in Figure 10.

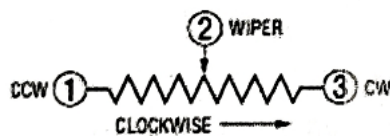


Figure 10. schematic symbol

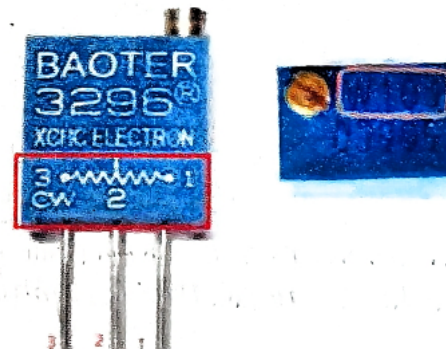


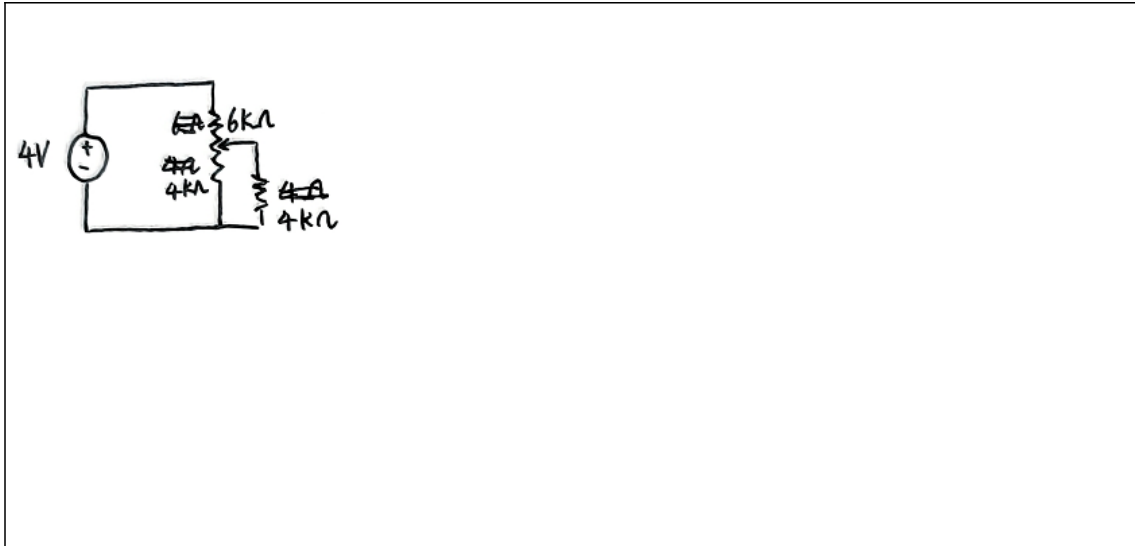
Figure 11. 10kΩ potentiometer with labelled terminals

There are different types of trimmer pots available in market such as through hole, single and multi-turn variation. A potentiometer has a sliding or rotating knob which is called a wiper. The resistance between terminals T1-T3 (R_{13}) is fixed ($R_{13} = X\Omega$ in the $X\Omega$ potentiometer) whereas the resistance between T1-T2 (R_{12}) and T2-T3 (R_{23}) changes based on the position of the wiper.

A W103 ($10 \times 10^3 = 10\text{k}\Omega$) potentiometer is shown in figure 11. As for a 10kΩ potentiometer, it means that the maximum resistance of the variable resistor is 10kΩ and

the minimum is 0Ω . This means that by changing the wiper position, you get a value between 0Ω and $10k\Omega$.

Now, construct a voltage divider which is equivalent to the circuit in Figure 5 by using a potentiometer W103 Potentiometer.(6 pts)



PART EIGHT

Read the document about using Signal Generator.

Appendix4_ Signal Generator

Appendix 5_ Digital Oscilloscope

Appendix6_SDS2000X Digital Oscilloscope user guide

Appendix9_SDG2000X Signal Generator user guide

1. What is the main purpose of Function Generator?(3 pts)

A Function Generator is an electronic instrument that generates various types of periodic ~~electro~~ electrical signals.

2. What is the main purpose of an Oscilloscope? (3 pts)

An Oscilloscope is an ~~electri~~ electronic measurement instrument that visually displays the waveform of ~~electro~~ electrical signals over time on its screen.